GraphPathology enables geometry-aware reconstruction of spatial transcriptomics from sparse molecular measurements

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Abstract

Reconstructing spatial transcriptomics from limited molecular measurements requires accounting for where observations are collected. GraphPathology combines observed molecular anchors with spatial representations of tissue morphology. We investigated different forms of graph neural networks (GNNs). In two breast-cancer cohorts comprising 25 evaluation patients, reconstruction from 20% distributed molecular observations achieved median patient Pearson correlations of 0.359 and 0.416. We then examined measurement placement in 22 previously evaluated patients, holding the hidden locations, reconstruction method and number of observed sites fixed. Excluding anchors from a central region reduced median paired patient correlation by 0.154 and increased mean squared error by 45.0% for inverse-distance interpolation; all 22 patients deteriorated. Across matched external comparisons, the tested GNN predictors did not consistently improve reconstruction over the same-information molecular baseline. These findings identify anchor placement as an important determinant of sparse reconstruction for the specified sampling design and delimit the contribution of morphology-guided modeling under cross-cohort conditions. Measuring one fifth of locations offers a conditional opportunity to reduce location-dependent costs; total assay savings require prospective validation. Code: https://github.com/HySonLab/GraphPathology.

Keywords: spatial transcriptomics; sparse reconstruction; geometry-aware graph learning; spatial transcriptomics imputation; histopathology; breast cancer

1 Introduction

Spatial transcriptomics (ST) links expression measurements to tissue organization[1], but dense profiling can limit study scale. Sparse measurement designs raise two related questions: how the placement of measured sites affects reconstruction, and whether histology supplies useful information beyond the molecular observations already available. Answering these questions requires separating the information supplied to a predictor from the spatial design on which it is evaluated.

We study reconstruction of a fixed gene panel within a tissue section. Complete panel measurements are available at observed anchors; expression at hidden sites is withheld, while their coordinates and histology remain known. This setting differs from predicting an entirely

unassayed panel or inferring expression from images alone. Breast-cancer ST studies provide paired data for examining these distinctions across tissue sections and platforms[2–5].

1.1 Related work

Histology-based prediction ranges from local convolutional methods such as ST-Net[6] and DeepSpaCE[7] to models with contextual representations. Hist2ST combines Transformer and graph components[8]; THItoGene[9] and FmH2ST[10] further examine visual context and feature-informed links between distant regions. CMRCNet[11] and cross-cohort benchmarking[12] highlight the importance of transportability. Graph convolutional networks (GCNs)[13], GraphSAGE[14], GraphGPS[15] and equivariant graph networks[16] offer complementary representation principles; their application does not by itself establish a new biological interaction or a reconstruction advantage.

Some methods combine measured expression with images, including XFuse[17], TESLA[18] and iStar[19]. Sparse-acquisition and tissue-wide mapping systems include Sparser2Sparse[20], SCR^2-ST[21] and PanoSpace[22], with differing modalities and sampling assumptions. SPACEL addresses spatial-domain and multi-section tasks[23]. Their training requirements, target resolutions and available inputs differ from ours. We therefore do not infer superiority or equivalence to these systems from comparisons with a frozen image predictor or simple molecular interpolation.

1.2 Framework and study objectives

GraphPathology distinguishes two graph roles. A coordinate-defined tissue-context graph aggregates histological representations, whereas an observation-dependent molecular-support graph connects hidden sites to measured anchors. The latter transports measured expression and supplies the strongest external reconstruction signal in this study. Keeping these roles separate permits same-information comparisons of molecular estimators and tests of the incremental contribution of image context.

We evaluate distributed reconstruction across two breast-cancer studies, isolate anchor placement in a controlled 22-patient comparison with identical hidden targets and equal anchor counts, and investigate several GNN representations against molecular and conventional neural baselines. Coverage, contiguous missing regions and distance-resolved errors characterize failure regimes. The purpose is to establish what the available measurements support, including negative results, rather than to equate architectural complexity with novelty or measured-location reduction with total assay savings.

2 Results

2.1 Study design separated information access and dataset roles

The task used a fixed 50-gene panel and paired spatial expression and histology (Fig. 1; Table 1). Four HEST patients supplied 35,536 locations: two patients for fitting, one for validation and one for development evaluation. Eight HER2ST patients (3,455 locations) supported transportability and calibration development; their reuse precludes treating them as

untouched external validation. The original external evaluation comprised three held-out Wu patients and the first numbered section from each of 22 Bassiouni patients, totaling 25 patients and 38,484 locations. All 50 genes were retained externally; HER2ST used a fixed 40-gene intersection.

The additional placement controls and neural comparisons reused these already evaluated cohorts and one development patient. They are exploratory follow-up analyses, not new blind replication. Image-only predictors receive histology and permitted geometry, molecular estimators receive coordinates and same-section anchor vectors, and fusion or residual predictors receive both. Thus image-only versus molecular comparisons quantify the value of measured anchors under unequal information access; incremental image value requires comparison with the same anchors supplied to both methods. The public external images were downsampled, containing only 15/25 native pixels in Wu and 24–29 in Bassiouni across the nominal 112-μm patch.

Table 1: Cohorts and their analytical roles. Counts refer to retained locations and patient-level study roles, not independent spot replicates. The same eight HER2ST patients were first inspected for transportability and subsequently used for calibration development. The Wu study contains 15,601 locations in all six patients; only 8,481 locations from its three evaluation patients enter the original external evaluation. All 22 Bassiouni patients were evaluated without new training or calibration. "Pixels per patch" refers to the native image information in a nominal 112-μm field of view, before resizing. The primary panel was fixed in development.

Cohort	Platform	Patients, total / evaluation	Locations, total / evaluation	Genes	Analytical role and image quality
Development (HEST)	Targeted Xenium	4 / 1	35,536 / 20,761	50/50	Two training, one validation, one primary evaluation patient; conditional four-patient sensitivity analyses. Estimated 0.21–0.36 μm/pixel.
HER2ST	Sequencing ST	8 / 8	3,455 / 3,455	40/50	Initial transportability, then calibration development on the same patients. Approximately 0.69–0.92 μm/pixel.
Wu	Visium	6 / 3	15,601 / 8,481	50/50	Three calibration and three original evaluation patients. Only 15 or 25 native pixels per patch.
Bassiouni	Visium	22 / 22	30,003 / 30,003	50/50	First numbered section per patient; original external evaluation with no cohort-specific fitting. Only 24–29 native pixels per patch.

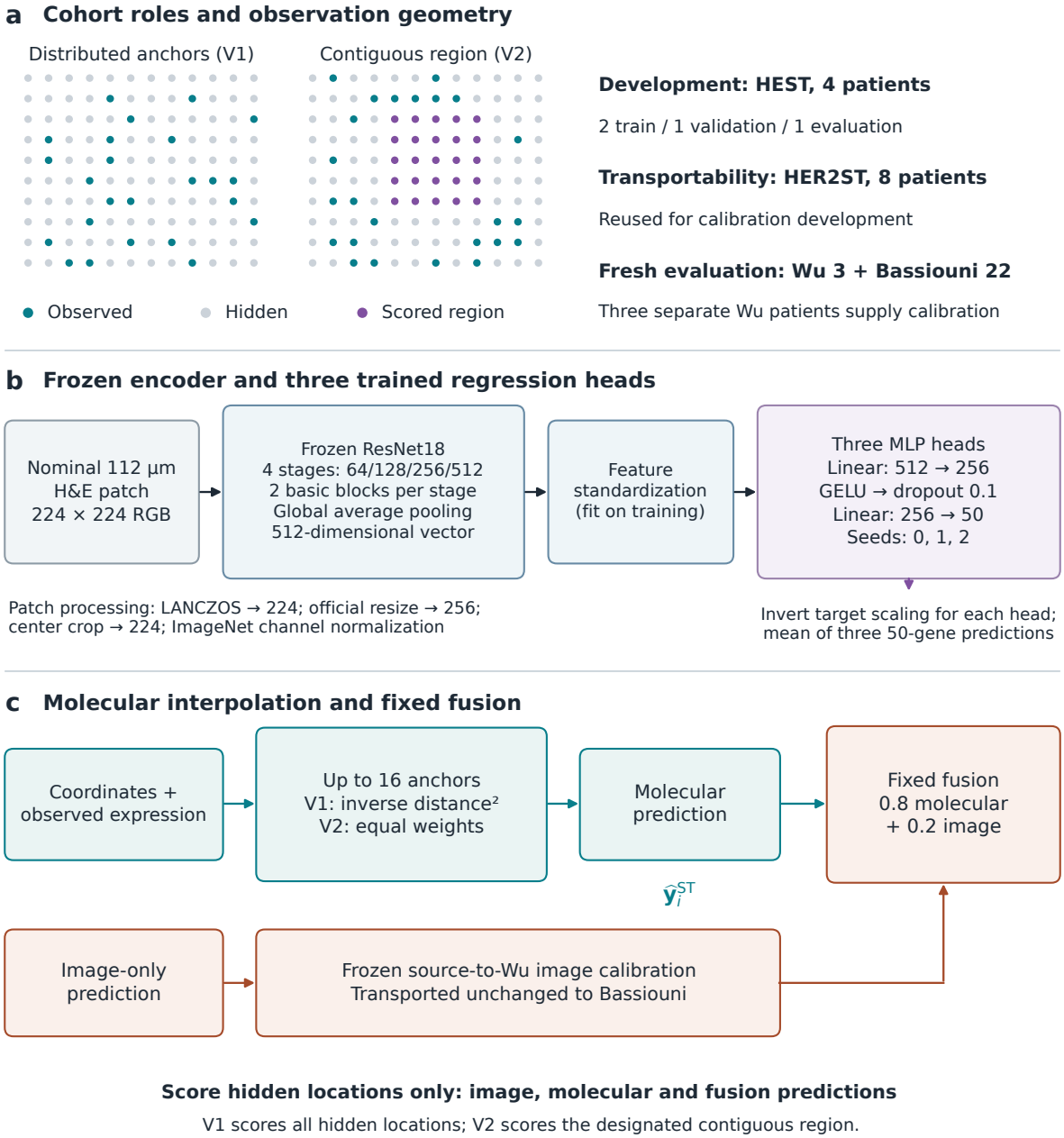


Figure 1: Study design and neural-image/molecular reconstruction architecture. a, Cohort roles and illustrative observation geometries. Teal locations are observed molecular anchors; gray locations are hidden; purple locations form the contiguous scored region. The synthetic spatial examples illustrate 20% coverage and are not patient measurements. The Wu calibration and evaluation groups contain different patients. **b,** The frozen ImageNet ResNet18 maps histology to a 512-dimensional vector after global average pooling; the ImageNet classifier is removed. External nominal 112-μm patches are first resized to 224×224 with LANCZOS, then processed by the official resize-to-256, central-crop-to-224 and channel-normalization transform. Feature scaling fitted on the training patients precedes each $512 \to 256 \to 50$ MLP, with GELU activation and dropout 0.1 after the hidden linear layer. The three heads use seeds 0, 1 and 2. Their predictions are inverse-scaled to the common target units and averaged. **c,** Molecular interpolation uses up to 16 observed neighbors, with inverse-distance-squared weighting for distributed evaluation (V1) and equal weighting for contiguous regional completion (V2). Fixed fusion combines this output with the calibrated image prediction using weights 0.8 and 0.2. The uncalibrated image ensemble is also evaluated separately. Calibration uses the development-training and Wu calibration patients and is transported unchanged to Bassiouni. Evaluation uses hidden locations only. V1 and V2 differ in weighting and scoring support as well as observation geometry.

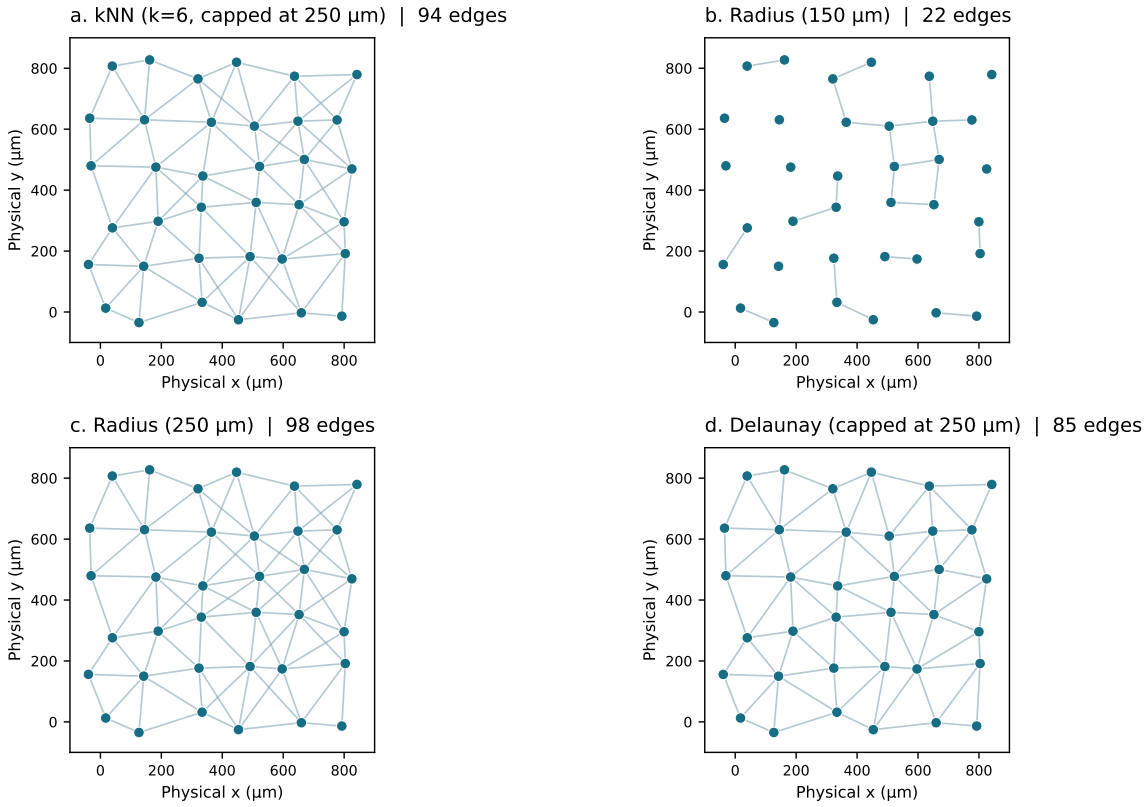


Figure 2: Geometry-aware graph construction and contextual representations. a–d, The same 36 synthetic tissue locations are connected by (a) a symmetrized six-nearest-neighbor graph with a 250-μm distance cap, (b,c) 150- and 250-μm radius neighborhoods, or (d) a 250-μm-capped Delaunay graph. Coordinates and axes are identical across panels and all retained edges are displayed. Symmetrization means that a node can have more than six neighbors. **e,** Geometry determines a fixed adjacency/aggregation operator acting on the frozen image embeddings before an expression-regression head. The molecular-reconstruction stage uses a separate observation-dependent graph joining hidden locations to up to 16 measured molecular anchors. Synthetic points illustrate the architecture only, not a fitted patient graph or a performance result.

2.2 Distributed molecular observations supported reconstruction across cohorts

At 20% distributed coverage in TENX99, median gene Pearson correlations were 0.695 for image-only prediction, 0.792 for interpolation and 0.822 for fixed fusion; corresponding mean squared errors (MSEs) were 0.911, 0.268 and 0.279 (Fig. 3; Table 2). The correlation increase from fusion therefore did not imply an error reduction. These are one-patient development results.

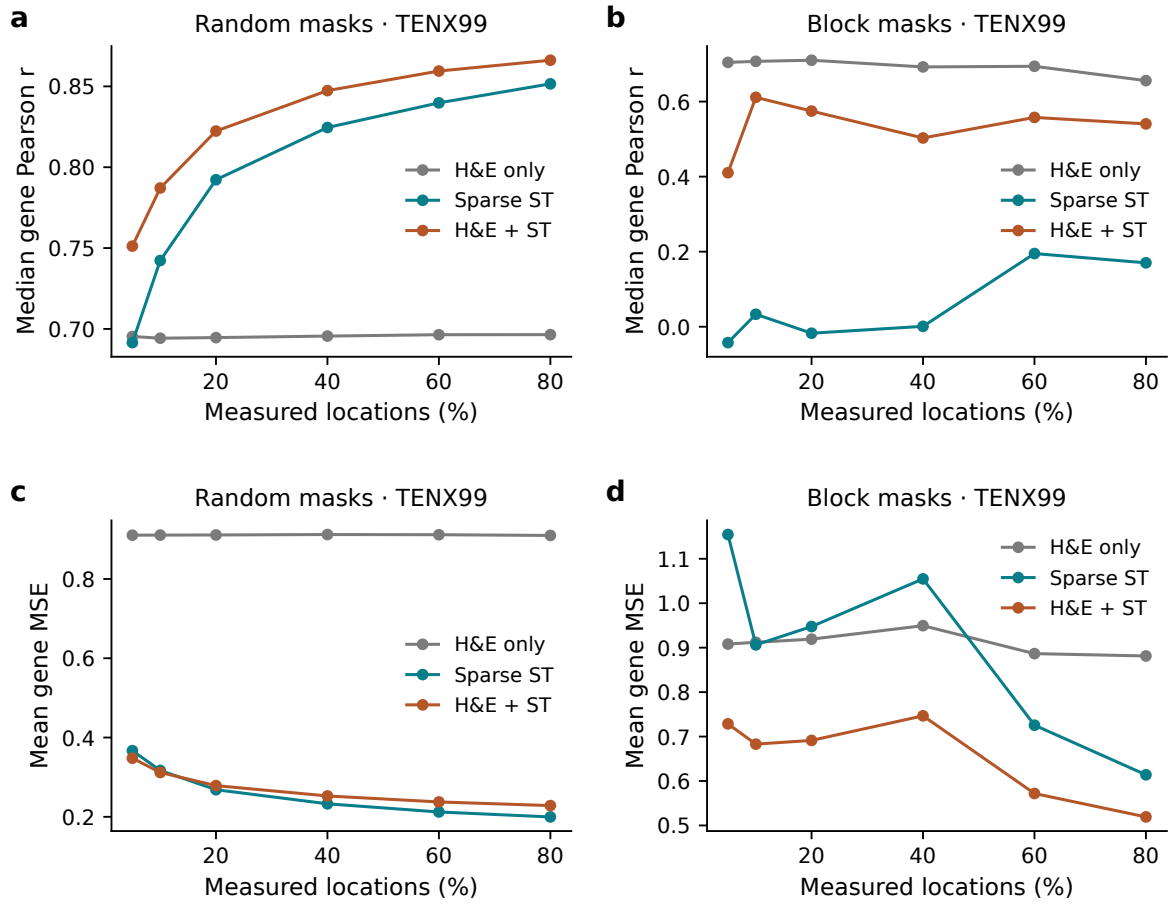


Figure 3: Development reconstruction depends on molecular coverage and masking regime. Results are from the single primary development patient, TENX99, using the frozen 50-gene panel and ResNet18 representation. **a,c,** Distributed random observation. **b,d,** The earlier development block-mask regime. Panels a and b show median gene Pearson correlation; panels c and d show mean gene MSE across measurement fractions. The image-only, selected molecular interpolation and fixed fusion curves summarize three frozen mask replicates per condition; curves are means over masks, not patient confidence intervals. Targets and error units are unchanged across methods. The development block masks differ from the external V2 forbidden-region protocol and should not be treated as directly equivalent geometry tests. At 20% distributed coverage, fusion has higher correlation than interpolation but slightly higher MSE, illustrating that spatial association and calibration are distinct properties. The image-only comparator receives no observed molecular anchors.

In the original Wu evaluation, the median of patient-level median gene correlations was 0.359 for interpolation and 0.059 for image-only prediction. Corresponding MSEs were 0.0987 and 2.9155, and mean absolute errors (MAEs) were 0.2179 and 1.5578 (Fig. 4). Bassiouni interpo-

lation achieved correlation 0.416, MSE 0.0923 and MAE 0.2033, compared with image-only values 0.075, 1.9673 and 1.2120 (Fig. 5). Every evaluation patient favored interpolation in correlation, MSE and MAE at each tested budget. All 22 Bassiouni patients met the predefined joint support criterion, which was descriptive rather than a significance test.

Table 2: Reconstruction at 20% distributed measurement coverage. Each entry is a cohort median of patient-level summaries after averaging mask seeds within geometry and geometries within patient, except that the development row represents the single primary patient. Pearson r is the median correlation across panel genes; MSE and MAE are patient means across genes and hidden locations before the cohort median. All methods within a cohort use identical target units and observation masks. Different gene panels and assay count distributions preclude treating these values as a pooled cross-cohort ranking. Fusion rules and their calibration are frozen but differ between the development and external protocols; see Methods and Supplementary Information. The image-only branch has no same-section molecular anchors.

Cohort	n	Genes	Method	Pearson r	MSE	MAE
Development	1	50	Image only	0.695	0.9110	0.7606
			Spatial interpolation	0.792	0.2684	0.3904
			Fixed fusion	0.822	0.2787	0.4173
HER2ST, zero-shot	8	40	Image only	0.143	4.3665	1.9376
			Spatial interpolation	0.194	0.0680	0.1730
			Fixed fusion	0.255	0.3481	0.5153
Wu, original	3	50	Image only	0.059	2.9155	1.5578
			Spatial interpolation	0.359	0.0987	0.2179
			Fixed fusion	0.335	0.1002	0.2321
Bassiouni, original	22	50	Image only	0.075	1.9673	1.2120
			Spatial interpolation	0.416	0.0923	0.2033
			Fixed fusion	0.407	0.1022	0.2288

Increasing distributed coverage from 10% to 40% improved correlation and MSE in every patient. Bassiouni correlations at 10/20/40% were 0.380/0.416/0.465, with MSEs 0.1001/0.0923/0.0852; Wu values were 0.313/0.359/0.393 and 0.1080/0.0987/0.0899 (Supplementary Table S9). These budget-specific protocols use nested anchor sets but different hidden sets. They neither identify a universally sufficient budget nor compare an identical target set across coverages. Earlier sampling comparisons, including label-free spatial coverage sampling, are retained in Supplementary Fig. S8.

2.3 Controlled anchor placement altered reconstruction in all 22 patients

The controlled experiment fixed a central region R and a hidden subset H, then changed only the eligible placement of an equal number of anchors. Distributed anchors could occupy any non-H location; region-excluded anchors were restricted outside R. Within each comparison, the estimator, gene panel, hidden targets and total anchor count were identical. This design isolates placement for the specified central-region sampling rule.

At 20% coverage, region exclusion worsened Pearson, MSE and MAE in all 22 Bassiouni patients for both fixed estimators (Fig. 6; Table 3). For inverse-distance weighting (IDW), the median paired correlation change was −0.1541 (descriptive 95% patient-bootstrap inter-

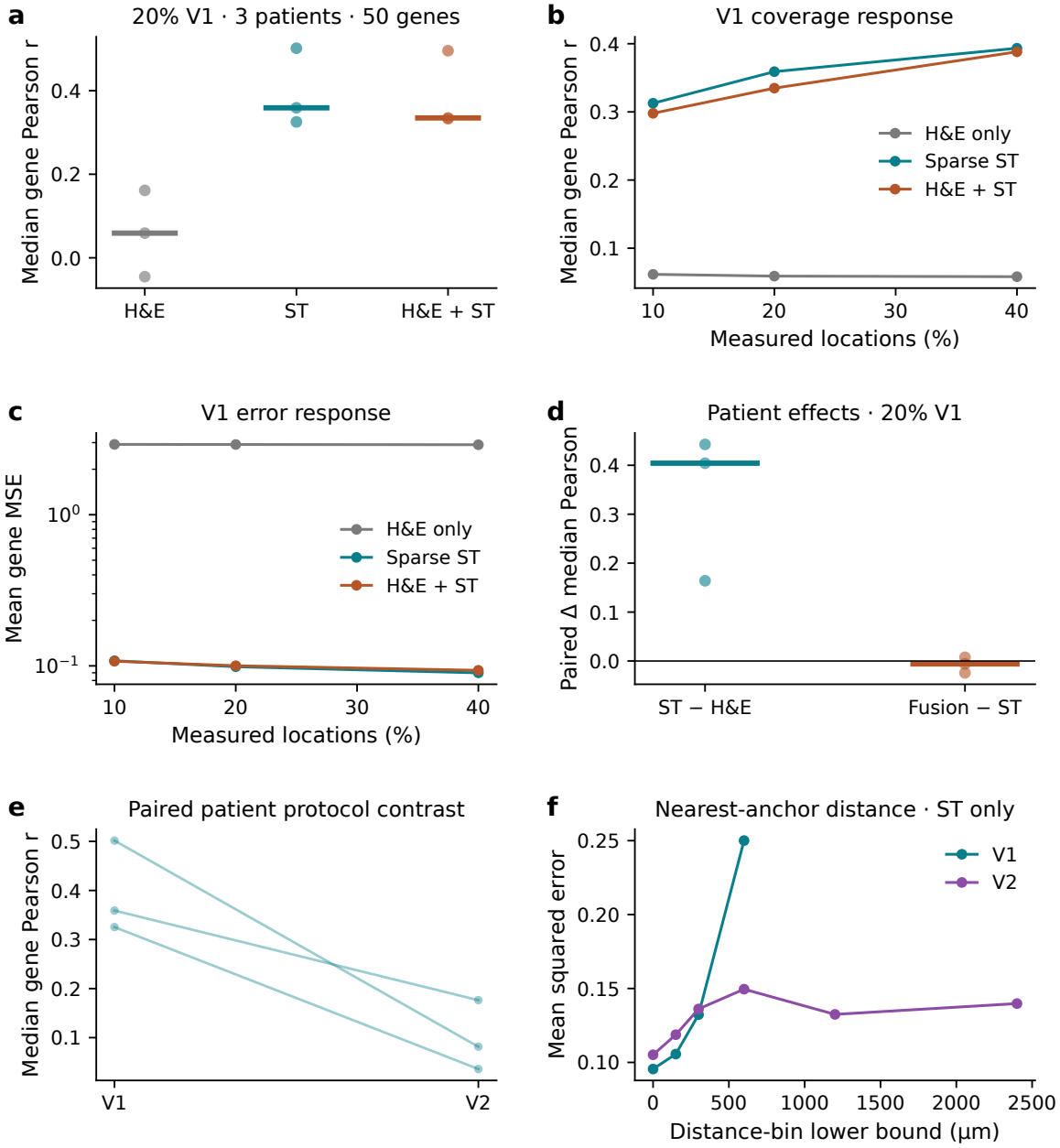


Figure 4: Fresh evaluation in three held-out Wu patients supports interpolation but not robust image fusion. The frozen 50-gene panel was evaluated in patients 1160920F, CID4290 and CID44971, none of whom contributed to Wu calibration. **a,** Patient-level correlation summaries at 20% distributed coverage. **b,c,** Median patient correlation and MSE across 10%, 20% and 40% distributed coverage. **d,** Paired patient correlation effects for interpolation versus image-only prediction and fusion versus interpolation. **e,** Paired distributed (V1) and contiguous-region (V2) molecular correlations at 20%. **f,** Distance-resolved molecular MSE at 20%. Patient summaries average ten random mask seeds within geometry and then geometry means; cohort summaries are medians, not pooled spot estimates. Three fixed contiguous geometries are used for V2. Each patient is a biological replicate; seeds are not. V1 uses inverse-distance-squared interpolation and V2 uses a nearest-anchor mean, so panel e is a protocol contrast. The public image inputs contained only 15 or 25 native pixels per 112-μm patch, limiting conclusions about morphology. Error units are common development-standardized log1p counts.

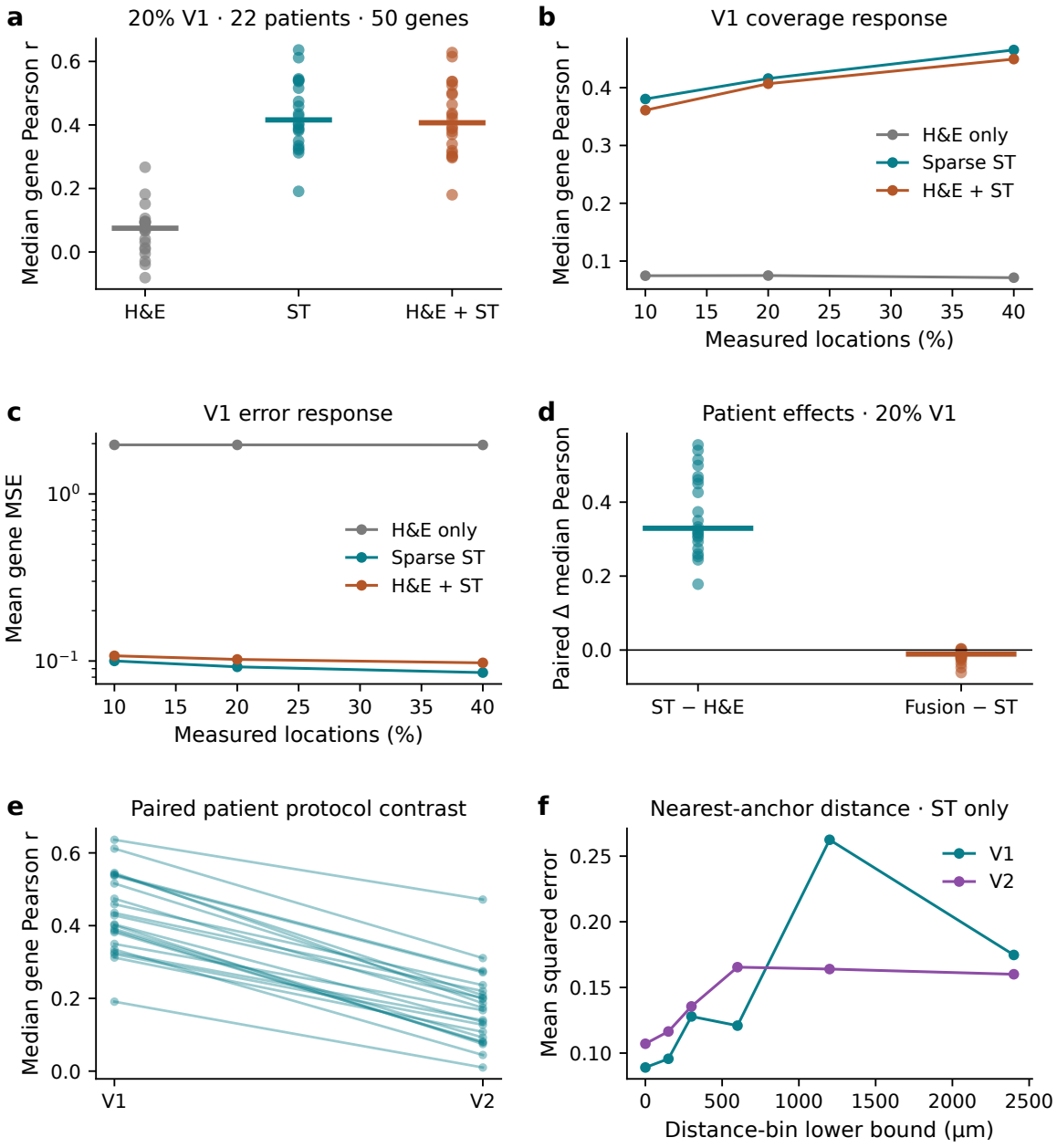


Figure 5: Independent 22-patient replication confirms the coverage response and weaker contiguous completion. One first-numbered section per Bassiouni patient was selected before evaluation, yielding 30,003 locations and the same 50 primary genes. No patient from this study was used for new training or calibration. **a,** Patient correlation summaries at 20% distributed coverage. **b,c,** Correlation and MSE across measurement fractions. **d,** Paired interpolation–image and fusion–interpolation correlation effects. **e,** Paired V1/V2 molecular correlations at 20%. **f,** Distance-resolved molecular MSE at 20%. Each patient summary averages ten mask seeds within geometry and then the three V2 geometry means where applicable; cohort curves use patient medians. All 22 patients improved in correlation and MSE between 10% and 40% V1 coverage and deteriorated in correlation, MSE and MAE under V2 relative to V1 at 20%. V1/V2 differ in scoring support and interpolation weighting as well as missing-region design. Image fusion uses the frozen Wu calibration and 0.2 image weight. Public histology contained only 24–29 native pixels per 112-μm patch. Distance curves have varying occupied-bin support and are not calibrated uncertainty intervals. No p-values or patient confidence intervals are implied.

val −0.2056 to −0.1096), relative MSE increased 45.0% (29.8–52.2%) and MAE increased 0.0335 (0.0266–0.0508). Uniform interpolation gave a correlation change of −0.1502 (−0.1805 to −0.1094), relative MSE increase 39.9% (29.2–53.5%) and MAE increase 0.0326 (0.0234–0.0462). All three budgets and all patient effects are retained. These intervals describe uncertainty across the 22 patients; they are not p-values or evidence of a universal geometry effect.

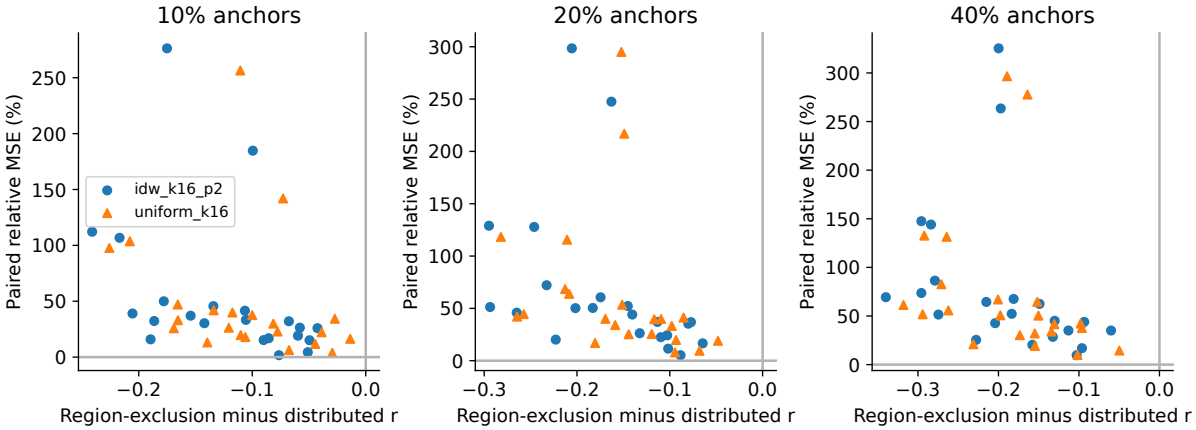


Figure 6: Identical-target placement effects in all 22 patients. Each point is one patient, averaged over ten mask seeds; shapes distinguish fixed IDW and uniform k16. Both anchor placements score the same H with equal anchor counts. Panels retain all three budgets. These are not historical V2-minus-V1 effects.

Table 3: Controlled Bassiouni placement effects. Region-exclusion minus distributed anchors at 20% coverage, identical H and anchor counts. Median paired patient effects with descriptive 95% percentile intervals from 10,000 patient resamples. Positive error changes deteriorate. All 22 patients are retained; masks are averaged within patient. The full 20-estimator and three-budget exports accompany the paper.

Operator	Effect	Median [95% interval]
IDW	Δr	-0.1541 [-0.2056, -0.1096]
IDW	MSE%	+44.9864 [+29.8116, +52.2347]
IDW	Δ MAE	+0.0335 [+0.0266, +0.0508]
Uniform	Δr	-0.1502 [-0.1805, -0.1094]
Uniform	MSE%	+39.9128 [+29.2387, +53.4974]
Uniform	Δ MAE	+0.0326 [+0.0234, +0.0462]

The three previously evaluated Wu patients corroborated the direction: at 20% coverage, region exclusion reduced IDW correlation by 0.106–0.245 and increased relative MSE by 20.1–45.4%; uniform reconstruction lost 0.093–0.188 in correlation and increased MSE by 21.7–38.9% (Fig. 7b; Supplementary Table S12). Each patient is reported individually. The original distributed-versus-regional protocol comparison is distinct: it changes geometry, estimator and scored support, rather than placement alone.

2.4 Matched spatial baselines separated estimator choice from sampling design

We compared twenty fixed molecular estimators on identical distributed masks in Wu and Bassiouni at each of three coverage levels: 10%, 20% and 40%. The grid included nearest-anchor, uniform, IDW and Gaussian reconstruction; all declared estimators are shown, without choosing an outcome-selected winner (Fig. 8; Supplementary Table S10 and source tables). Separately, IDW and uniform interpolation were crossed with the original distributed and regional masks (Fig. 7a; Supplementary Table S11). This separates the effect of changing the estimator from the controlled placement comparison, while retaining the original protocol-level results. Predetermined gene maps include poorly reconstructed and nearly constant fields (Supplementary Figs. S13,S14,S26 and S27).

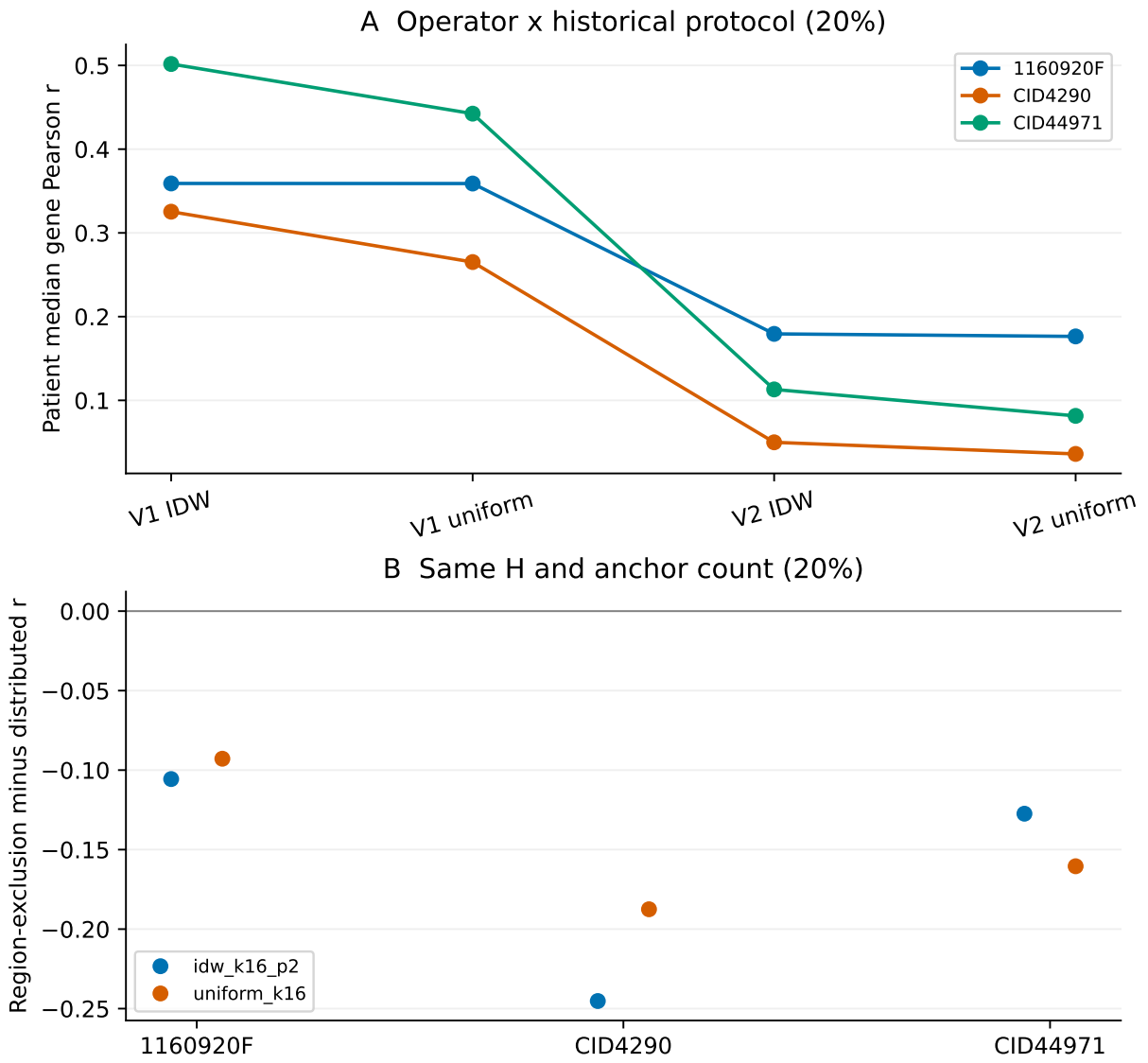


Figure 7: Matched estimator and placement controls in previously inspected Wu patients. (a) Both operators share each historical mask and scored set; V2 values average the three geometries. (b) Region-exclusion minus distributed correlation on the identical H, with equal total anchor count and ten random seeds. The three individual patients are shown; masks and genes are not biological replicates. All points use 20% coverage and the frozen 50-gene panel.

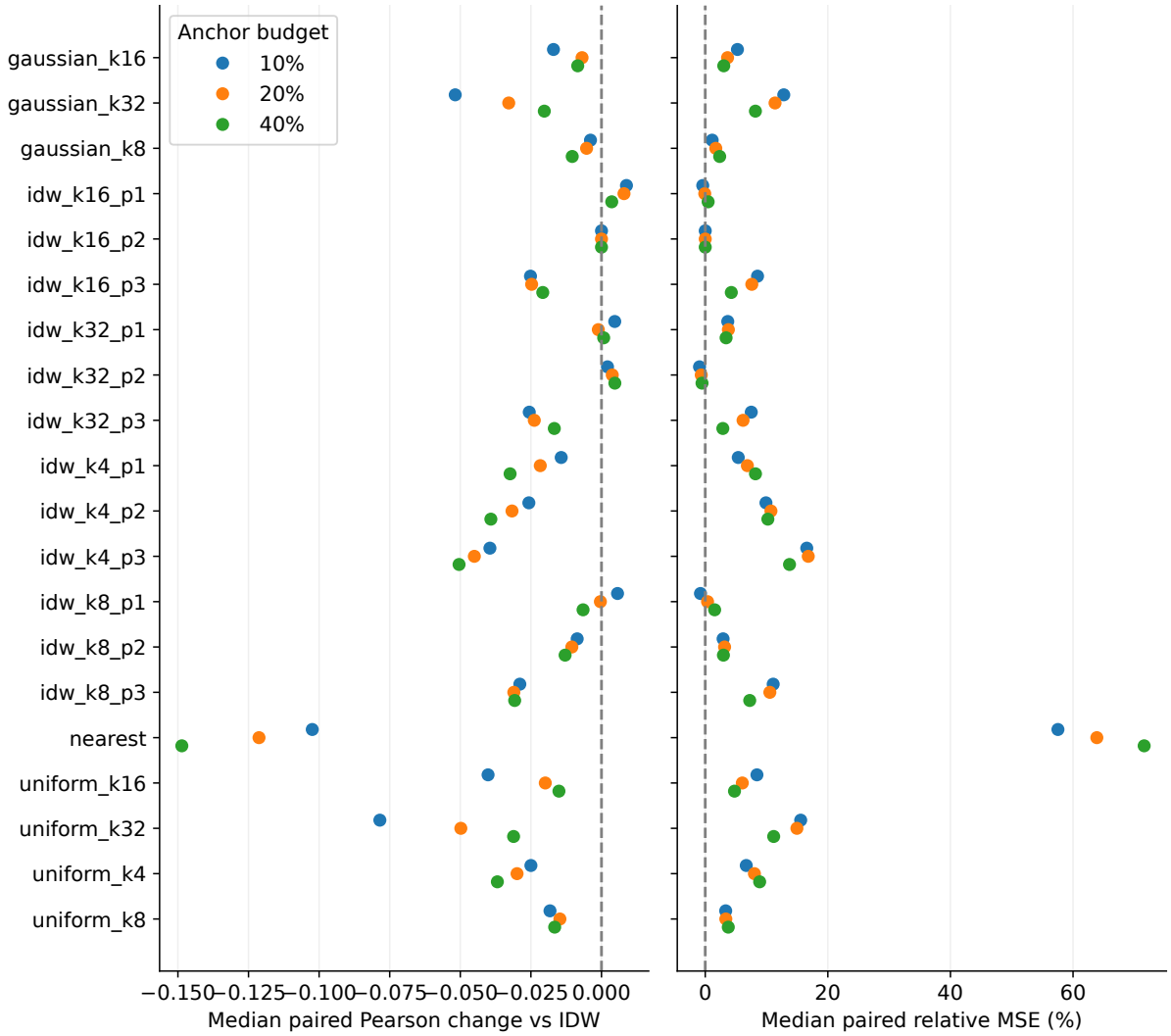


Figure 8: Complete matched-information baseline grid in 22 previously inspected Bassiouni patients. Every declared estimator and budget is shown. Points are medians of paired patient effects relative to fixed IDW k16/p2 after averaging ten original mask trajectories. Correlation increases and error decreases improve. No estimator was selected from these outcomes.

2.5 Graph representations did not add a consistent external benefit

The GNN comparisons included local multilayer perceptrons, fixed mean/GCN/GraphSAGE representations, bounded nonlocal models and fixed-distance or latent-coordinate message passing. Their architectural and optimization regimes are distinguished in Methods. In the 30-head minibatch development analysis, image-only seed-mean correlations for local, GCN and GraphSAGE predictors were 0.682, 0.654 and 0.678, with MSEs 0.9114, 0.8999 and 0.8859. Residual-anchor correlations were 0.821, 0.810 and 0.825, with MSEs 0.2567, 0.2510 and 0.2338; the identical-mask IDW reference was 0.792 and 0.2684. These one-patient results did not establish a topology-specific advantage (Fig. 9; Supplementary Table S13). The full seed comparisons, including shuffled controls, remain available.

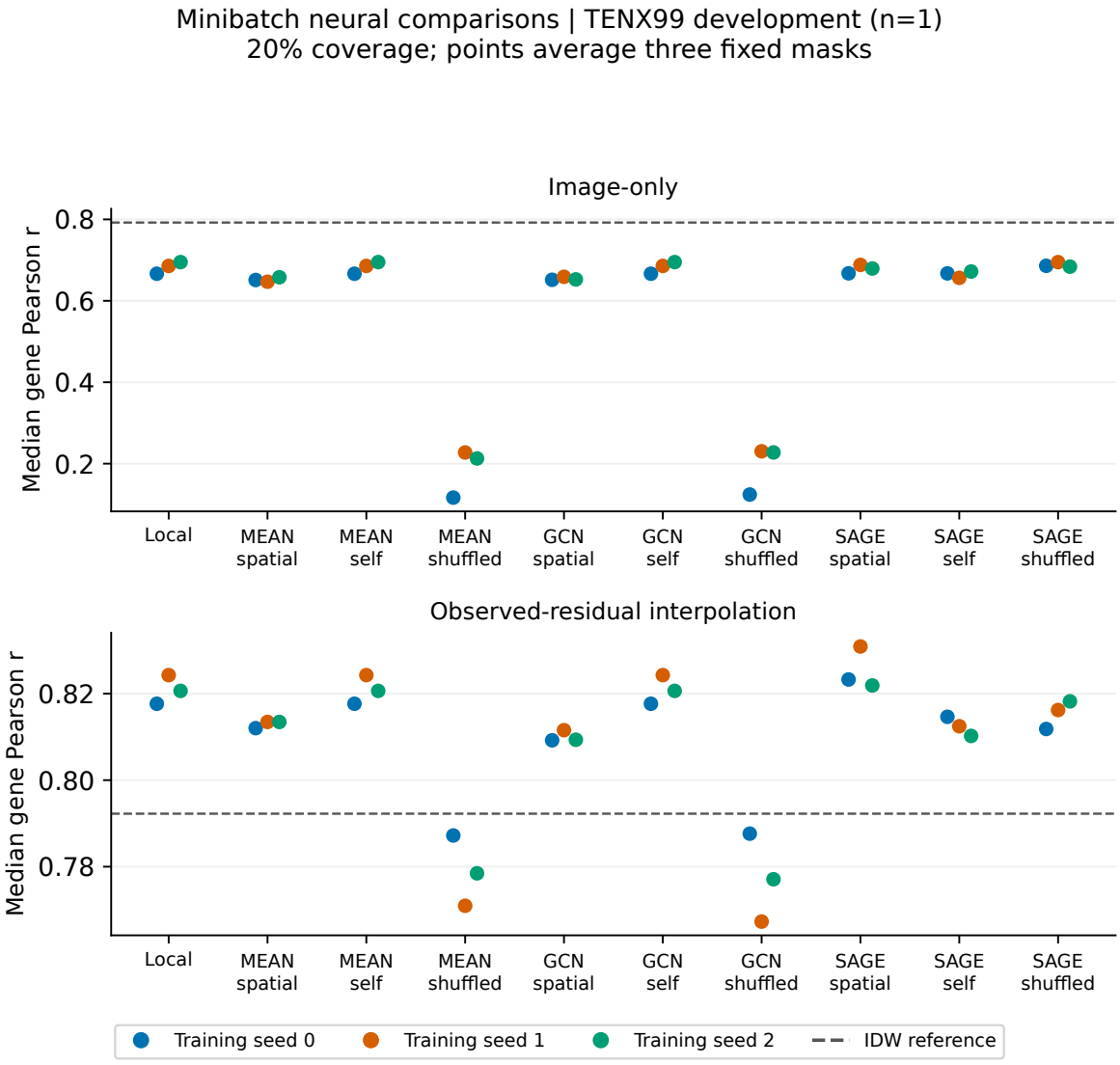


Figure 9: Minibatch neural comparisons in the development patient. Three training seeds for each local, mean, GCN and GraphSAGE configuration, including self-only and node-shuffled controls; all evaluated on TENX99, a single previously used development patient. Points average three fixed masks at 20% coverage. Dashed lines give identical-mask IDW reference values. Top: image-only prediction; bottom: the protocol-specified exploratory residual-anchor baseline. Color identifies training seed, not an independent patient. Panels use different y-ranges. IDW receives measured expression unavailable to the image-only branch; residual versus IDW tests incremental morphology. These fits were not tuned on Wu or Bassiouni outcomes.

The expanded matrix compared all 48 full-section-trained fits with the 30 frozen minibatch fits in TENX99, Wu 3 and Bassiouni 22, using identical masks and information routes (Fig. 10). In the primary 20% distributed Bassiouni comparison, the residual-anchor GraphGPS-inspired model changed paired patient correlation versus IDW by −0.0504 (95% interval −0.0629 to −0.0389), relative MSE by +13.4% (11.3–18.4%) and MAE by +0.0244 (0.0204–0.0295). The latent-coordinate EGNN gave −0.0480 (−0.0698 to −0.0368), +13.0% (9.1–18.3%) and +0.0252 (0.0188–0.0310), respectively. Neither primary model improved any of these endpoints in any of the 22 patients. Under identical-H region exclusion, their median relative MSE increases were 10.2% and 9.2% (Fig. 11).

Component comparisons also failed to establish the proposed gains: the full nonlocal model worsened correlation and errors relative to its local counterpart, and latent-coordinate updates did not improve the fixed-distance model (Table 4). Same-information conventional neural comparisons, all routes, individual Wu effects, training seeds and distance-bin support are retained in Supplementary Figs. S15–S25,S28 and Tables S14–S17. Conditional geometric and reach checks confirm the implemented properties; they do not establish biological long-range association or full-image equivariance. These negative findings concern the tested training budget, frozen features and low-resolution cross-cohort inputs.

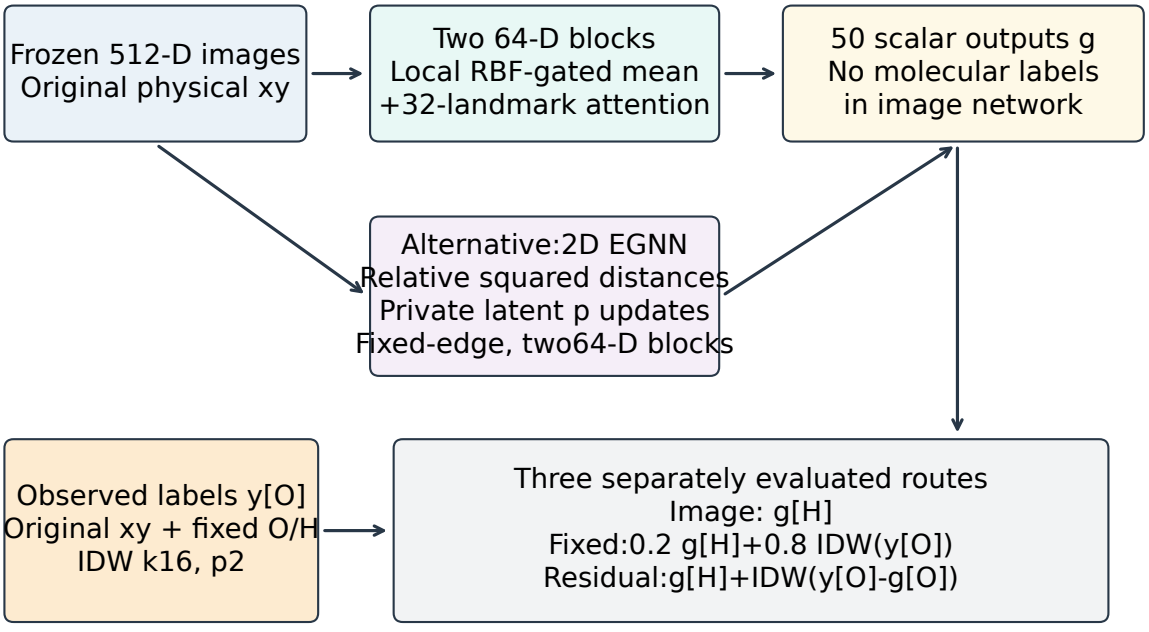


Figure 10: Graph predictors and information boundaries. The image network receives frozen image embeddings and coordinate-defined structure. The GraphGPS-inspired branch combines local gated aggregation with bounded attention to 32 coordinate-only landmarks. The alternative EGNN changes private latent coordinates through relative displacements; the distance-only model fixes them. Original physical coordinates always define masks, anchor interpolation and physical-distance diagnostics. No hidden expression is an image-network input.

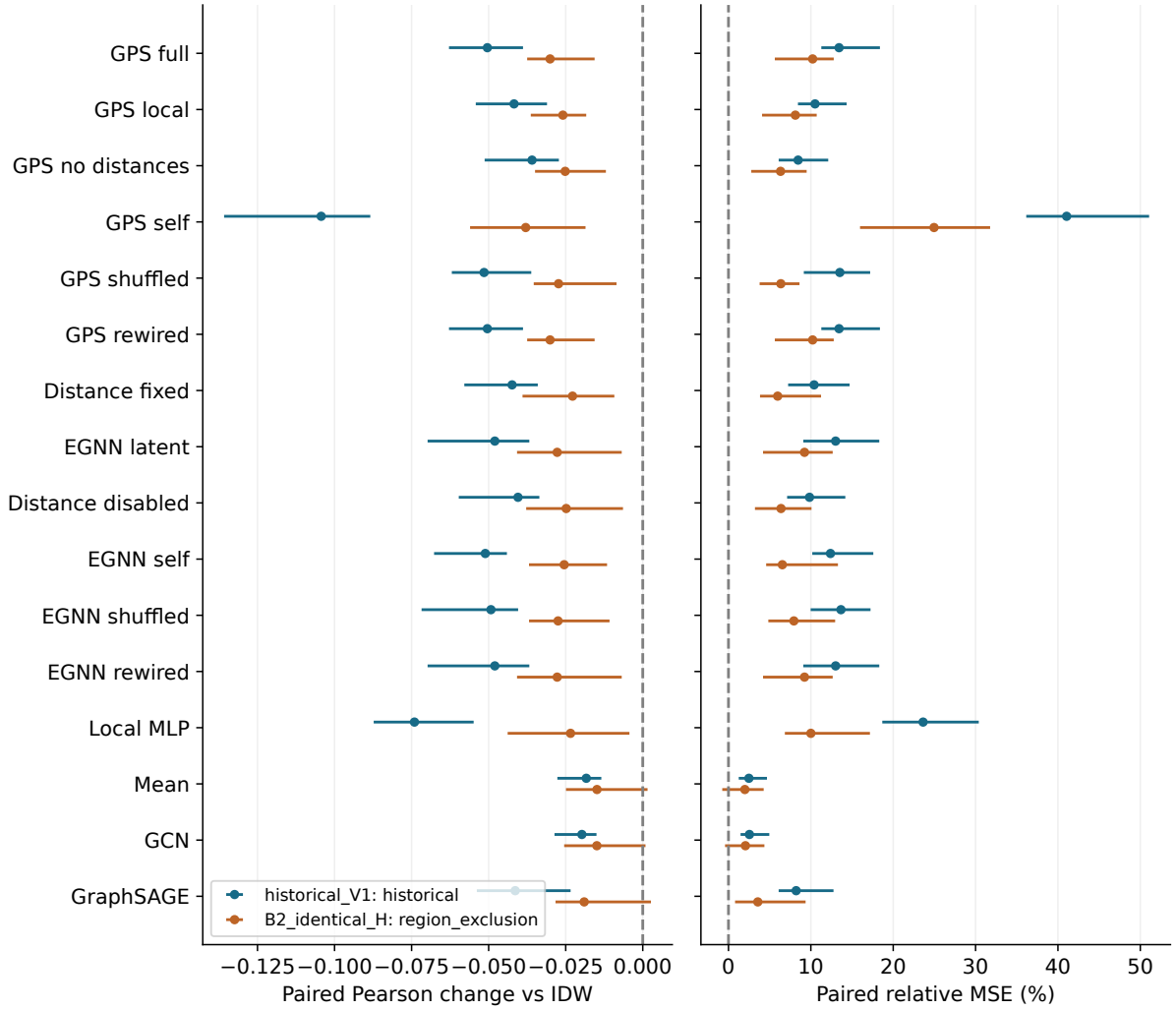


Figure 11: All 16 new architectures/controls versus molecular IDW in 22 Bassiouni patients. Exploratory residual route at 20% coverage; colors distinguish historical distributed V1 and identical-H region exclusion. Bars are descriptive 95% patient percentile intervals for median paired effects after mask/training-seed averages. Positive correlation and negative relative MSE improve. The two designs have different scored support and are not subtracted here. All models are shown regardless of direction; nominally equal parameter counts do not imply equal effective function classes.

Table 4: Predefined architecture contrasts in recovered Bassiouni patients. At20% historical V1 coverage, the exploratory residual route uses identical measured anchors for both models. Effects average all three training seeds and three mask seeds within each patient before pairing; relative MSE is normalized by the reference patient mean MSE after these averages. Intervals resample22 whole patients, not genes, masks or training seeds. They are descriptive, unadjusted intervals, not simultaneous significance claims. Other routes, controls, budgets and individual Wu effects are fully exported.

Candidate vs control	Effect	Median [95% interval]
GPS full vs GPS local	Δr	-0.0050 [-0.0074, -0.0044]
GPS full vs GPS local	MSE%	+2.7228 [+2.0494, +3.6169]
GPS full vs GPS local	Δ MAE	+0.0039 [+0.0030, +0.0054]
GPS full vs GPS no distances	Δr	-0.0121 [-0.0135, -0.0088]
GPS full vs GPS no distances	MSE%	+4.9138 [+3.6647, +5.7686]
GPS full vs GPS no distances	Δ MAE	+0.0064 [+0.0056, +0.0078]
EGNN latent vs Distance fixed	Δr	-0.0048 [-0.0074, -0.0008]
EGNN latent vs Distance fixed	MSE%	+1.6673 [+0.7107, +2.8664]
EGNN latent vs Distance fixed	Δ MAE	+0.0026 [+0.0014, +0.0046]
EGNN latent vs Distance disabled	Δr	-0.0065 [-0.0096, -0.0024]
EGNN latent vs Distance disabled	MSE%	+1.9736 [+1.0100, +3.3448]
EGNN latent vs Distance disabled	Δ MAE	+0.0028 [+0.0020, +0.0053]

2.6 Regional completion, distance and calibration defined failure regimes

The original contiguous-completion protocol was substantially harder than distributed reconstruction. At 20% coverage, Bassiouni correlation fell from 0.416 to 0.171, MSE increased from 0.0923 to 0.1496 and MAE from 0.2033 to 0.2624. Every patient deteriorated; median paired correlation change was -0.267 and relative MSE increase 52.8%. Wu correlation fell from 0.359 to 0.0815, with lower correlation and higher errors in every patient (Supplementary Table S9). Because the interpolation rule and scored tissue also change, these effects must not be attributed solely to geometry. A withheld region may still lie inside the anchors' convex hull.

Distance-resolved errors were heterogeneous (Fig. 12). At 20% Bassiouni coverage, farthest-bin MSE exceeded nearest-bin MSE in 15/22 patients under distributed interpolation and 20/22 under regional completion; strictly increasing curves occurred in only 6 and 13 patients. Wu distributed curves increased across occupied bins, while regional curves were nonmonotonic. Bin support and tissue composition differ, precluding a universal reliability cutoff (Supplementary Table S7).

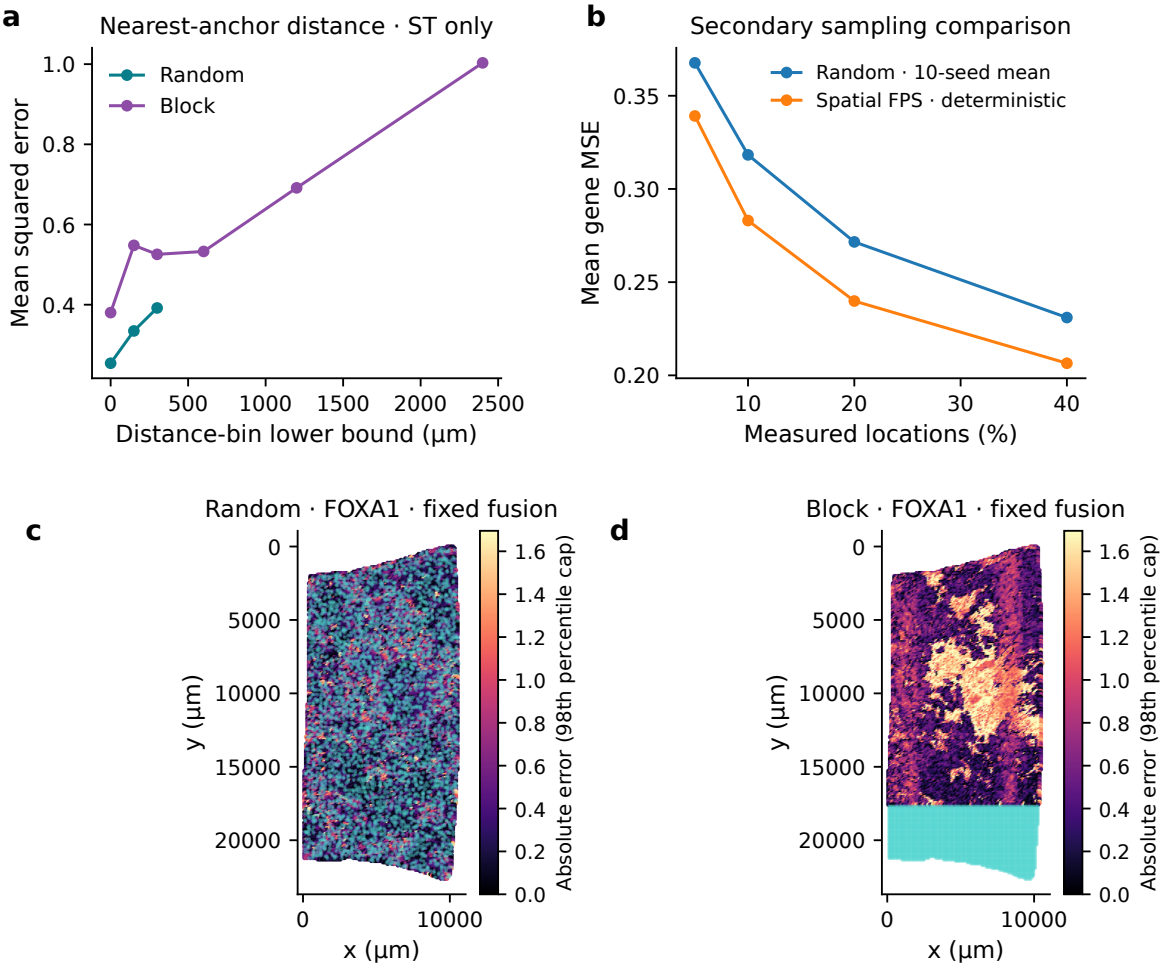


Figure 12: Spatial diagnostics reveal heterogeneous error beyond an overall prediction score. Development results use TENX99 and the frozen 50-gene panel. **a,** Mean gene MSE versus nearest-anchor distance for molecular interpolation at 20% coverage under distributed and development block masks; only occupied distance bins are shown. **b,** Development comparison of random and spatial farthest-point sampling, shown as MSE at matched measured-location budgets. These are descriptive single-patient sampling diagnostics, not a demonstration of universally optimal acquisition. **c,d,** Absolute-error maps for the preselected gene *FOXA1*, fixed fusion, 20% coverage and replicate 0 under random and block masks. Both maps use the same error scale, capped at the 98th percentile of the pooled preselected map-source values, which also include the archived capture-window condition; no locations are removed. Missing-region shape, available anchors and scored tissue differ between the maps. The figure illustrates why distance-bin support and spatial errors should accompany global correlation and why development geometry findings require independent evaluation.

Fixed image fusion also failed to improve external reconstruction consistently. Bassiouni correlation, MSE and MAE at 20% distributed coverage were 0.407, 0.1022 and 0.2288, versus 0.416, 0.0923 and 0.2033 for interpolation. Only 3/22 patients improved in correlation and none in either error. The median paired correlation effect was −0.0108 (95% interval −0.0187 to −0.0081), with relative MSE increase 10.7% (6.6–14.5%). Wu showed correlation improvement in 1/3 patients and no error improvement. The original anchor-versus-image paired correlation gain was 0.329 (0.304–0.438), a contrast with different information access (Table 5). HER2ST calibration reduced scale-related error but did not establish robust joint correlation/error gains (Supplementary Figs. S1,S9 and S10). An affine transform followed by normalized interpolation and its inverse left molecular predictions unchanged, as expected algebraically (Supplementary Methods).

Table 5: Patient-level uncertainty reanalysis of original Bassiouni exports. Median paired effects at 20% V1 coverage, with 10,000 patient-resampling percentile 95% intervals (seed 20261101). Negative error changes improve; positive correlation changes improve. These are descriptive exploratory intervals, not simultaneous bands or p-values. Anchors versus image is an unequal-information contrast; fusion versus anchors tests incremental image contribution. All original point estimates and source bytes are unchanged.

Contrast	Effect	Median	95% interval	Improved
Anchors minus image	Δr	0.3293	[0.3044, 0.4382]	22/22
Anchors minus image	Relative MSE (%)	-95.4534	[-96.2444, -94.0201]	22/22
Anchors minus image	Δ MAE	-1.0074	[-1.1341, -0.9108]	22/22
Fusion minus anchors	Δr	-0.0108	[-0.0187, -0.0081]	3/22
Fusion minus anchors	Relative MSE (%)	10.7078	[6.6170, 14.5048]	0/22
Fusion minus anchors	Δ MAE	0.0238	[0.0171, 0.0278]	0/22

Anchor-vector and anchor-position permutations degraded molecular reconstruction in predefined controls; image permutation supplied no comparable evidence for a useful image contribution (Supplementary Figs. S11 and S12; Table S4). These related controls and exact hidden-label invariance checks support computational traceability without supplying additional biological replicates.

3 Discussion

Measurement placement was a determinant of reconstruction quality even when the amount of observed expression was held fixed. The 22-patient comparison shows that equal budgets can produce different performance when anchors are excluded from the region containing the scored targets. Its strength is the fixed hidden set and estimator, which separate placement from changes in tissue support or reconstruction rule. The result applies to the specified central-region design; it does not establish a universally optimal sampling policy or biological correlation length.

Distributed molecular anchors nevertheless supported useful reconstruction of the predefined panel across two breast-cancer studies. The coverage response suggests that nearby same-section measurements carry information that a transported image predictor may not recover. This is a task-specific advantage of measured expression, not an equal-information demonstration that interpolation outperforms all image-based systems. Reliability should be assessed

on the intended acquisition design: favorable distributed masking does not ensure accurate completion of a large unmeasured region[24].

The neural negative results sharpen this distinction. With the same anchors supplied to residual predictors and molecular baselines, the tested graph representations did not produce a consistent external gain. Adding nonlocal access or updating latent coordinates did not establish the hypothesized advantages. Functional attention reach and conditional equivariance are software properties, rather than evidence of learned molecular relationships. These findings argue for retaining strong molecular comparators and explicit component controls, while avoiding a general rejection of geometry-aware learning on better-supported data.

Cross-cohort histology remains a substantial limitation. Public images were heavily downsampled, encoders were frozen, and fitting used only two source patients. Differences in capture depth, support and composition remain after mathematical normalization. Calibration can change image predictions, but cannot create a new interpolation benefit when a per-gene affine transform is inverted around the same normalized weighted average. Native-resolution representations, broader source training and genuinely independent evaluation would be needed to establish whether morphology adds reproducible incremental information.

The scope is also limited by retrospective masking, a development-selected 50-gene panel (40 genes in HER2ST), small Wu patient numbers and restriction to breast tissue. The exploratory controls reused evaluated cohorts. Genes, sites and repeated masks do not increase biological sample size, and descriptive patient intervals do not establish clinical generalizability. Distance alone did not define a reliable error threshold; occupied-bin support and patient-specific failures should remain visible before reconstructed expression is used downstream. Approaches such as TISSUE motivate explicit uncertainty assessment[25], but calibrated predictive uncertainty was not evaluated here.

Coverage is a count of measured locations, not a measured financial saving. If a full assay has fixed cost C_0 and per-location cost c for N locations, the conditional ratio is

$$\frac{C_{\text{full}}}{C_{20\%}} = \frac{C_0 + cN}{C_0 + 0.2cN} \leq 5. \tag{1}$$

Only a purely location-dependent component could fall fivefold at 20% coverage. Selective capture, laboratory expenditure, sequencing requirements and downstream accuracy were not prospectively tested. Future acquisition studies should evaluate placement and error requirements together with realized costs.

GraphPathology therefore supports an information-explicit account of sparse reconstruction: measured molecular support can be useful, its placement matters under controlled conditions, and added image or graph complexity requires a demonstrated incremental benefit. Prospective sampling and independent native-resolution cohorts are the next tests of practical value.

4 Methods

4.1 Data, target panel and preprocessing

All experiments used previously released paired spatial expression and histology data. The four development samples were obtained through HEST[5]: NCBI783 and NCBI785 were the

training patients, TENX95 was the validation patient and TENX99 was the primary development evaluation patient. The cohort comprised 7,015 training locations, 7,760 validation locations and 20,761 primary evaluation locations. Earlier conditional leave-one-patient-out analyses reused frozen gene panels and development-selected configurations and were interpreted as sensitivity analyses, not unbiased external cross-validation.

HER2ST[2] contributed one patient-specific section for each of A1–H1. Initial evaluation was followed by patient-disjoint calibration development using the same inspected cohort; these two analytical roles are distinguished throughout. In the Wu local Visium cohort[3], patients 1142243F, CID4465 and CID4535 supplied calibration statistics, and patients 1160920F, CID4290 and CID44971 remained held out. The Bassiouni evaluation cohort was obtained from GSE210616[4]. The source study includes 43 sections from 22 patients. We selected the first numbered section for every published patient from metadata before target evaluation, yielding BASS_P01–P09, P11–P15 and P17–P24. Gaps in identifiers reflect source numbering rather than outcome-dependent exclusion. The source study's own reference and validation groups were not used as calibration groups in this reconstruction study.

Metadata, repository identifiers, prior-use records and available asset hashes were checked to exclude reused or reprocessed development samples. Such checks support cohort independence but cannot conclusively establish patient identity across unrelated anonymized repositories. Cohort roles and sample counts are summarized in Table 1; identifiers and exact panels are supplied in Supplementary Tables S1 and S2.

The primary panel contained the 50 genes ranked using development-training expression variance; ranking did not use evaluation expression. The development assay provided 280 harmonized biological genes, and larger 250- and 280-gene settings were used in earlier development analyses. The original external studies retained all 50 genes. HER2ST retained the frozen 40-gene intersection without replacing missing genes. The optional 278/280 intersection in the Bassiouni metadata was not used to claim a secondary whole-transcriptome experiment.

For the second replication, the source filtered feature-barcode H5 files, tissue positions, scalefactor JSON files and tissue images were acquired for each selected section. Barcode joins and published in-tissue flags defined included locations. No new expression-derived quality threshold, tissue-class annotation or deconvolution output was used to select evaluation locations. Counts were checked for finite, nonnegative integer values. For count c_{ig} at location i and gene g, the evaluation target was

$$y_{ig} = \frac{\log(1 + c_{ig}) - \mu_g^{\mathrm{dev}}}{\sigma_g^{\mathrm{dev}}}, \tag{2}$$

where the centering and scaling constants were the pooled location mean and population standard deviation frozen from development-training patients; a standard deviation below 10^{-8} was replaced by 1. No destination library-size or capture-area correction was fitted. Identical target units were used for all methods within each comparison. These common mathematical units do not imply cross-platform calibration of absolute molecular abundance. Supplementary Fig. S2 summarizes cross-cohort patterns without pooling panels; the gene-level results are provided in Supplementary Table S3.

4.2 Registration and frozen image features

Visium array coordinates were mapped to image coordinates by a least-squares fit to the source hexagonal lattice with 100-μm spacing. Coordinate residuals, image scale and anisotropy were checked against the frozen qualification settings, and overlays were inspected before outcome evaluation. These tests concern location consistency and do not establish cell-level registration. Images were not super-resolved.

The Wu and Bassiouni images contained only 15/25 and 24–29 native pixels, respectively, across the nominal 112-μm patch width. Resizing does not recover morphology absent from these public inputs. Image quality was therefore recorded as a limitation of the image comparator, not as a restriction on spatial interpolation that uses only coordinates and observed expression.

At each external location, an RGB patch covered a nominal 112-μm square, rounded to the available native pixels. The patch was resized with LANCZOS to 224×224 pixels. The torchvision `IMAGENET1K_V1` preprocessing then resized the shorter side to 256 pixels using bilinear interpolation, took a central 224×224 crop, rescaled intensities to $[0, 1]$, and normalized RGB channels using means $(0.485, 0.456, 0.406)$ and standard deviations $(0.229, 0.224, 0.225)$. The second resize and central crop were retained explicitly; 112 μm denotes the nominal extracted patch, rather than the final uncropped field seen by the network. Development embeddings used the supplied 224×224 HEST patches with the same torchvision transform. No stain normalization, random image augmentation or encoder fine-tuning was applied.

The frozen ResNet18[26] comprised a 7×7 convolutional stem with stride 2, batch normalization, a rectified-linear activation and 3×3 max pooling with stride 2, followed by four residual stages. The stages contained two basic blocks each, with channel widths 64, 128, 256 and 512. Each block used two 3×3 convolutions with batch normalization and a rectified-linear activation after the first convolution and after shortcut addition. Identity or projected shortcuts matched the feature dimensions. Global average pooling produced a 512-dimensional vector; the original ImageNet classifier was replaced by the identity operation. All encoder parameters and batch-normalization statistics remained fixed during feature extraction.

The final central crop retains approximately 0.875 of the extracted patch linear field.

4.3 Coordinate-defined graph representations

Each retained location is a node with physical position $q_i \in \mathbb{R}^2$ in micrometers and a frozen image embedding. Following message-passing principles[27], graphs were constructed independently within each section from physical coordinates, without expression labels. The candidates were symmetrized six-nearest-neighbor adjacency capped at 250 μm, radius graphs at 150 or 250 μm, and Delaunay adjacency capped at 250 μm. Symmetrization takes the union of directed nearest-neighbor edges, so final degree need not equal six. The binary adjacency A had no self edges. With $\widetilde{A} = A + I$, degree matrix $\widetilde{D}$ of $\widetilde{A}$, and degree matrix D of A, the three fixed operators were

$$P = \widetilde{D}^{-1}\widetilde{A}, \qquad Q = \widetilde{D}^{-1/2}\widetilde{A}\widetilde{D}^{-1/2}, \qquad R = D^{-1}A.$$

For R, an isolated node has a zero neighbor aggregate. The input to the MLP was PX for mean pooling, QX for GCN and $[X, RX]$ for mean-GraphSAGE, where X contains standardized frozen image embeddings and brackets denote concatenation. Each predictor therefore applies one fixed graph aggregation followed by the one-hidden-layer MLP defined above. These single-step predictors used neither learned adjacency nor attention; the additional message-passing models are defined below.

Graph choice was selected at seed 0 by the highest validation median gene Pearson correlation, then lower MSE and deterministic graph-name order. All three 50-gene predictors selected capped Delaunay adjacency. Mean pooling and GCN used width 128; GraphSAGE used width 67 to approximately match the initial local head's parameter count after doubling the input dimension. With ResNet18, these models contain 72,114 and 72,075 trainable head parameters, respectively; all use dropout 0.1. Supplementary Table S8 summarizes the selected dimensions.

4.4 Neural predictors and distinct training regimes

All neural fitting used NCBI783/NCBI785 and validation-only stopping on TENX95. The following regimes are distinct; parameter similarity does not imply identical update schedules or effective capacity.

Original primary image ensemble. Each expression head was a one-hidden-layer MLP with the exact sequence

$$512 \xrightarrow{\text{linear}} 256 \xrightarrow{\text{GELU}} 256 \xrightarrow{\text{dropout (0.1)}} 256 \xrightarrow{\text{linear}} 50, \tag{3}$$

where GELU denotes the Gaussian error linear unit. Both linear layers included biases and used the default PyTorch initialization, uniform on $[-1/\sqrt{d_{\text{in}}}, 1/\sqrt{d_{\text{in}}}]$. The output was linear, with no nonnegativity constraint. Each head contained 144,178 trainable parameters. Feature-wise centering and scaling, S_x, were fitted to the pooled 7,015 training locations from NCBI783 and NCBI785. A separate per-gene scaler, S_y, was fitted to their already normalized expression targets. Both used population standard deviations, replacing scales below 10^{-8} with 1. If f_{θ_s} denotes the head trained with seed $s \in \{0, 1, 2\}$ and ϕ_{R18} the frozen encoder, the image prediction in the common target units was

$$\widehat{\boldsymbol{y}}_i^{\text{image}} = \frac{1}{3} \sum_{s=0}^{2} S_y^{-1}[f_{\theta_s}(S_x(\phi_{\text{R18}}(x_i)))] \,. \tag{4}$$

Dropout was disabled at inference. The three inverse-scaled predictions were averaged without fitting ensemble weights.

Heads minimized mean squared error in the additionally standardized target space using AdamW, learning rate 10^{-3}, weight decay 10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. Weight decay applied to all head parameters, including biases. Each epoch shuffled the training locations into minibatches of 512, retaining the final partial batch, for at most 80 epochs. Training used CPU float32 computation and no learning-rate scheduler. Early stopping used the TENX95 validation patient: an improvement required validation MSE to decrease by more than 10^{-4}, and training stopped after 10 consecutive epochs without such an improvement.

The best qualifying parameters were restored. Hidden widths $\{128, 256\}$ and dropout probabilities $\{0, 0.1\}$ were compared at seed 0; the configuration with the highest validation median gene Pearson correlation was selected, with lower validation MSE breaking ties. This configuration-selection criterion was distinct from the within-fit stopping criterion. The selected width-256, dropout-0.1 heads were fitted with seeds 0, 1 and 2; their retained epochs were 12, 11 and 12, respectively. No external patient was used to select or refit these heads.

Additional minibatch heads. Thirty additional local, mean, GCN and GraphSAGE heads (three seeds per specified configuration/control) used the following settings. They were fitted separately from the original primary ensemble and were later frozen as references. A single learned linear/GELU/dropout/readout layer followed mean or symmetric GCN aggregation; GraphSAGE concatenated self and neighbor mean. Widths were 128 for local/mean/GCN and 67 for SAGE (72,114 versus 72,075 parameters). Spatial, self-only and degree-preserving node-shuffled controls shared each family's nominal parameters, labels and training settings. SAGE self-only concatenated X with X and has a smaller effective function class. AdamW used lr 10^{-3}, decay 10^{-4}, batch size 512, dropout 0.1, maximum 80 epochs, patience 10 and minimum validation improvement 10^{-4}; TENX95 alone selected stopping. Seeds 0/1/2 were all retained. Deterministic float32 CPU execution used four torch/BLAS threads; Local, mean-self and GCN-self identity controls used the same inputs. Their float32 weight/prediction comparisons allowed rtol=atol=$32\epsilon_{\text{float32}} = 3.8147 \times 10^{-6}$; this tolerance did not relax float64 molecular replay or exact hidden-label invariance. Further identity diagnostics are in Supplementary Methods.

Full-section comparison. The 48 additional fits crossed 16 configurations with seeds 0/1/2. AdamW used learning rate 10^{-3}, weight decay 10^{-4}, maximum 80 epochs, patience 10 and minimum validation-MSE improvement 10^{-4}. Each epoch made one full-section update per source training patient in seeded random order. Local/mean/GCN widths were 128 and concatenated GraphSAGE width 67. These baselines match the new models' update budget, whereas the 30 frozen minibatch heads do not. Both additional regimes trained directly in the common development target units, without the primary ensemble's extra head-target scaler. Model selection retained every seed; TENX99 was used only for evaluation. Parameter counts, selected epochs, histories and computational costs are in Supplementary Tables S15 and S16. Initial development searches, Phikon representation comparisons[28] and conditional leave-one-patient-out fits have separate specifications in Supplementary Methods S6.

4.5 Bounded nonlocal and distance-aware message passing

This implementation draws on GraphGPS principles rather than reproducing the reference implementation. Its blocks use a 512-to-64 projection, two 64-wide blocks, four attention heads, dropout 0.1, pre-LayerNorm, local linear messages gated by eight relative-distance radial features, parallel bounded attention, residual additions and a two-layer GELU feed-forward path. The local graph is Delaunay with a 250 μm cap. Global keys/values are at most 32 deterministic coordinate-only farthest-point landmarks, giving $O(NL)$ attention storage with $L \leq 32$; there is no dense N-by-N allocation. The nonlocal-disabled model attends self plus up to seven nearest graph neighbors, padding unused slots with self. Thus full/local changes attention

range and key count as well as connectivity, despite identical 121,018 nominal parameters. Local radial features use log(1+d/250) with eight centers 0–1 and width 0.15; attention relative bias uses eight centers 0–4 and width 0.5. The distance-disabled version removes explicit gates/biases but retains coordinate-defined graph/landmark selection and is not coordinate-free.

The invariant fixed-coordinate and EGNN alternatives use two 64-wide message blocks. With centered normalized $p_i = (q_i - \bar{q})/250$, where q_i is the physical position in micrometers, $m_{ij} = \phi_e(h_i, h_j, \mathrm{RBF}(\|p_i - p_j\|^2))$, node updates aggregate the mean message through ϕ_h, using SiLU, LayerNorm and dropout 0.1. Eight radial centers span 0–4 with width 0.5. EGNN additionally updates private coordinates as

$$p_i^{l+1} = p_i^l + \frac{1}{\max(1, |\mathcal{N}_i|)} \sum_{j \in \mathcal{N}_i} 0.1 \tanh(\phi_x(m_{ij}))(p_i^l - p_j^l).$$

Each ϕ_x is a bias-free 64-to-1 map initialized at standard deviation 0.001. Only the first coordinate update affects the second layer's scalar readout; the last update has no downstream scalar loss. EGNN has 87,282 nominal parameters versus 87,154 for fixed distance (128 additional coordinate coefficients, of which the last 64 do not affect prediction). Both are E(2)-compatible conditional on frozen features, so their accuracy difference cannot identify equivariance itself. Latent positions never change original masks, anchors or IDW distances.

Self-only and coordinate/graph-shuffled variants retain nominal parameter tensors but have different effective function classes; inactive attention or edge parameters do not make them capacity-equivalent. Conjugating the graph and assigned coordinates together preserves edge-length geometry while breaking image-to-location assignment. Degree/length rewiring preserves each node's degree and global 50 μm edge-length bins below the same 250 μm cap, targeting $E/4$ accepted double swaps with at most $20E$ attempts. Actual successes are reported, including infeasible or weak rewiring. No hidden expression enters any graph, landmark or distance encoding.

4.6 Observation masks and controlled placement

For a section with N retained locations, the observed set $\mathcal{O}$ contained $\lfloor Nb + 0.5 \rfloor$ anchors at budget $b \in \{0.10, 0.20, 0.40\}$. Every hidden location lacked its entire expression vector; coordinates and image features remained available. Primary random trajectories used seeds 0–9 with nested prefixes, so higher budgets added anchors without discarding earlier observations. All trajectories were frozen without using expression values.

In distributed interpolation (V1), anchors were eligible throughout the tissue and all unobserved locations were scored. In contiguous regional completion (V2), a forbidden region consisted of the nearest 40% of locations to a center determined from the 25th, 50th or 75th percentile of the first coordinate principal-component rank. The three centers defined three fixed geometries. Anchors were sampled outside the forbidden region, and only the forbidden region was scored. All other unobserved labels were also removed from model inputs. Thus the same nominal budget refers to the full retained section even when evaluation is restricted to a region.

A deterministic spatial farthest-point policy was retained as a secondary sampling analysis. It

started from the eligible location farthest from the eligible-coordinate centroid and selected locations without expression labels. It was not optimized on external outcomes. Earlier development block and capture-window masks differ from the final regional-completion definition and are labeled separately in Fig. 3 and the source tables.

The controlled identical-target placement experiment (B2) used only the predeclared central PCA-coordinate region R (40% of locations). H was a fixed random 20% subset of R (about 8% of all locations), selected using SHA 256("H|patient") first eight bytes, little-endian. Distributed anchors were drawn from all non-H sites; region-exclusion anchors from outside R. Seeds 0–9 gave nested prefixes with equal total counts at every budget. R, H, barcodes and anchor trajectories were persisted before scoring. Both estimators scored exactly H; no expression-based geometry or exclusion was used. Seed means preceded geometry means and patient pairing. Both primary error and correlation effects and nearest-anchor distances were exported, with occupied-bin errors and support.

The additional neural matrix used three original random masks per budget for TENX99, and original V1/central-V2 geometry 1 masks with seeds 0/1/2 externally. All 78 fits also used B2 seeds 0/1/2 at 10/20/40% budgets. This subset is distinct from the ten-seed, three-geometry molecular baseline analysis.

4.7 Molecular interpolation, calibration and residual prediction

For every unmeasured location $i \in \mathcal{U}$, we define directed support edges $i \to j$ to at most 16 measured locations $j \in \mathcal{O}$, identified by the smallest coordinate distance. The resulting observation-dependent edge-weight matrix $W_{\mathcal{U}\mathcal{O}}$ has row-normalized positive weights and zero entries for unavailable anchors. The molecular reconstruction operator is

$$\widehat{Y}^{\mathrm{ST}}_{\mathcal{U}} = W_{\mathcal{U}\mathcal{O}} Y_{\mathcal{O}}. \tag{5}$$

The operator has no explicit maximum-distance cutoff, permitting cross-gap transport of observed expression. It is not a learned attention matrix, and absence of a cutoff is not evidence of reliable long-range inference. The distinct V1 and V2 weights and scoring supports are defined next.

The molecular predictor used the up to 16 nearest observed anchors, $\mathcal{N}_{16}(i;\mathcal{O})$. For distinct observed and hidden coordinates, distributed interpolation used normalized inverse-distance-squared weights, a local form of inverse-distance interpolation[29]:

$$\widehat{y}^{\mathrm{ST}}_{ig} = \sum_{j \in \mathcal{N}_{16}(i;\mathcal{O})} w_{ij} y_{jg}, \qquad w_{ij} = \frac{\|q_i - q_j\|^{-2}}{\sum_{k \in \mathcal{N}_{16}(i;\mathcal{O})} \|q_i - q_k\|^{-2}}. \tag{6}$$

For the frozen V2 protocol, the same neighborhood size was used with equal weights $w_{ij} = 1/|\mathcal{N}_{16}|$. These rules were fixed during the Wu evaluation and were not selected on Bassiouni outcomes. Fewer than 16 anchors, if available, were handled by using all available anchors. Euclidean distances were rounded to six decimal places in micrometers and floored at 10^{-6} μm before inverse-square weighting; the normalized weights therefore remained finite at coincident coordinates.

The image-only prediction was denoted $\widehat{y}^{\mathrm{image}}$. The final fusion was

$$\widehat{y}_{ig}^{\mathrm{fusion}} = 0.8\widehat{y}_{ig}^{\mathrm{ST}} + 0.2\widetilde{y}_{ig}^{\mathrm{image}}, \tag{7}$$

where the image calibration was

$$\widetilde{y}_{ig}^{\mathrm{image}} = a_g^{\mathrm{W}} + b_g^{\mathrm{W}} \frac{\widehat{y}_{ig}^{\mathrm{image}} - a_g^{\mathrm{S}}}{b_g^{\mathrm{S}}}. \tag{8}$$

Here S denotes the two development-training patients and W the three Wu calibration patients. Moments were computed from all retained targets of the designated source and calibration patients, rather than their sparse anchor subsets or image predictions. Each patient contributed equal total weight: for a domain with P patients and n_p locations in patient p,

$$a_g = \frac{1}{P}\sum_{p=1}^{P}\frac{1}{n_p}\sum_{i=1}^{n_p} y_{pig}, \qquad b_g = \sqrt{\frac{1}{P}\sum_{p=1}^{P}\frac{1}{n_p}\sum_{i=1}^{n_p}(y_{pig} - a_g)^2}. \tag{9}$$

A scale below 10^{-8} was replaced by 1. This scale includes variation about the shared domain center; it is not the mean of patient-specific standard deviations. All moments used the common target units in Eq. 2. Bassiouni evaluation reused these source and Wu constants without fitting new moments or fusion weights. The calibrated image term used in fusion therefore differs from the raw image-only comparator reported in Table 2.

Per-gene affine transformation, normalized interpolation and inverse transformation are algebraically equivalent to interpolation in the original units (Supplementary Note S1). Calibration therefore changes the image term, not the underlying normalized molecular operator.

For additional graph comparisons, g_{ig} denotes the gene-g image prediction at site i in the common target space, calibrated by Eq. 8 for both external cohorts. Alongside image-only g and fixed fusion, we evaluated observed-residual interpolation,

$$\widehat{y}_{ig} = g_{ig} + \sum_{j\in\mathcal{O}} w_{ij}\,[y_{jg} - g_{jg}]. \tag{10}$$

Only observed residual vectors enter this predictor; no regression coefficients are learned from the evaluation section. It is distinct from the development ridge residual correction, which fits gene-specific coefficients to image and molecular graph summaries using training patients (Supplementary Methods S6). Neither should be interchanged with fixed image fusion. HER2ST patient-disjoint calibration and fusion-family selection are also specified in Supplementary Methods S6; the final external image weight and Wu-derived moments remained fixed.

The matched-estimator grid used all ten original random V1 trajectories at each of 10/20/40% coverage for nearest anchor; uniform k=4,8,16,32; IDW at those k with p=1,2,3; and Gaussian k=8,16,32. We retained cKDTree ordering, six-decimal distances and a 10^{-6} μm inverse-weight floor. Gaussian bandwidth was the median observed-anchor distance to its kth other observed neighbor, with k limited by available anchors; it never used hidden expression. Every declared estimator is reported. The operator-by-protocol comparison (B1) held each original mask and scored set fixed while changing IDW k16/p2 versus uniform k16, for both V1 and all three V2 geometries.

4.8 Metrics, patient effects and uncertainty

All expression metrics were computed on hidden locations only; the full observed expression vector was available at each anchor and the full target vector was withheld at every other location. Observed rows were selected before molecular prediction, so hidden expression could not influence anchor values or interpolation weights. For each gene, Pearson and Spearman correlations summarized spatial association, while MSE and MAE summarized prediction error. Panel Pearson values were medians across genes with defined correlations. Constant-target or constant-prediction correlations were left undefined rather than assigned zero; such genes did not pass positive-correlation threshold counts. Panel MSE and MAE were means of per-gene errors. In formulae, for the scored hidden set $\mathcal{H}$,

$$\mathrm{MSE} = \frac{1}{G|\mathcal{H}|}\sum_{g=1}^{G}\sum_{i\in\mathcal{H}}(\widehat{y}_{ig}-y_{ig})^2, \qquad \mathrm{MAE} = \frac{1}{G|\mathcal{H}|}\sum_{g=1}^{G}\sum_{i\in\mathcal{H}}|\widehat{y}_{ig}-y_{ig}|. \tag{11}$$

Within each patient and geometry, metrics were averaged over mask seeds; geometry means were then averaged equally. Headline cohort values were medians of these patient summaries. Tables label an MSE value as a cohort median of patient mean MSEs, not a pooled mean over all spots. Paired effects were calculated within patients before taking their median. This distinction is necessary because a median paired effect need not equal the difference or ratio of marginal medians.

The patient was the biological unit. Seeds described computational or mask variability, not additional patients. The original external-evaluation support criterion required a median gene Pearson gain of at least 0.02 over image-only prediction, relative mean MSE reduction of at least 2%, and non-worsening MAE. The cohort rule required at least $\lceil 2N/3 \rceil$ patients meeting the joint criterion at at least two budgets, positive median benefits at every budget, and the same support threshold for correlation/MSE improvement from 10% to 40%. For 22 patients this threshold was 15. The limited native image resolution restricts interpretation of the image comparisons even when the molecular support criterion is satisfied.

For the additional neural comparisons, mask means were formed within geometry, geometry means within patient and training seeds averaged within patient before paired effects. Relative MSE is the candidate-minus-reference patient mean MSE divided by the reference mean MSE after these averages. Bassiouni intervals resampled 22 paired patients 10,000 times (seed 20261101) and took percentile 95% limits for the median effect. They are descriptive, unadjusted intervals, not p-values or simultaneous bands. Cohorts were never pooled; Wu effects are individual, and no population interval was inferred from TENX99. Raw Spearman was secondary because near-tied ranks are sensitive to floating-point differences. Full defined-correlation counts accompany the panel summaries.

For each hidden location, the nearest-anchor distance was $d_i = \min_{j\in\mathcal{O}} \|q_i - q_j\|$. Distances were grouped into the frozen bins $[0, 150)$, $[150, 300)$, $[300, 600)$, $[600, 1200)$, $[1200, 2400)$ and $[2400, \infty)$ μm. Error summaries were reported for occupied bins; the Bassiouni binwise correlations required at least 20 scored locations, while the Wu distance summaries contained errors only. Empty bins were omitted, not filled with zero. Counts, geometry and run identities were retained to distinguish sparse support from low error. Nearest-versus-farthest contrasts and monotonicity summaries were descriptive properties of the occupied-bin curves, not estimates of a universal biological length scale.

4.9 Controls and reproducibility

In the second replication, control patient BASS_P01, 20% coverage, random-mask seed 0, V1 geometry 0 and V2 geometry 1 were designated before evaluation. Three control seeds permuted observed expression-vector assignments, anchor-position assignments or image predictions. The first two disrupt related molecular–location associations; they were not interpreted as independent biological evidence. Every saved prediction was also repeated after all unobserved expression was replaced with NaNs and with independent random values. Observed/hidden barcodes, mask identities, configurations and hashes accompanied the predictions.

Analyses used Python, NumPy[30], SciPy[31], pandas, PyTorch and related scientific libraries. Reproduction instructions and environment specifications are maintained with the code. Model training and original external evaluation used recorded configurations and checksums; source-data tables provide the numerical outputs underlying the published figures.

Graphs, anchor trajectories, feature scalers and landmark selection were label-free or fitted on the designated source patients only. Replacing every unobserved label, including unscored labels, with NaNs and independent random values had to leave fixed/residual predictions exactly unchanged. Image-only B2 metrics were also identical across placements and budgets because H was fixed. Input identities and barcode/coordinate joins were checked against original records; independently derived coordinates agreed with preserved label-free coordinates within $10^{-8}\,\mu$m, which governed exact neighbor ties. Numerical source tables, environment records and fitted-state hashes are retained in the repository; the detailed audit counts are summarized in the supplementary source guide.

The original external results were frozen before the additional controls. The latter followed timestamped local protocols, retained all specified configurations/seeds and reused previously inspected patients; they were exploratory, not formally preregistered. No evaluation outcome selected a new architecture, seed, calibration map or fusion coefficient. Original primary heads and newly fitted states have separate identities and availability records.

Conditional geometric tests applied rotations, reflections, translations and node permutations at float64/float32, recording scalar and physical latent errors separately. Float 64 graph-model tolerances were 10^{-8}; float32 and preaggregated baseline tolerances were 10^{-4}. Physical latent tolerances included the 250-μm normalization. Recomputed-graph tests used nondegenerate coordinates without distance ties. These tests condition on frozen image embeddings and documented graph/landmark handling, not invariance of the complete image pipeline. Attention diagnostics recorded physical/assigned connection fractions and actual mass beyond $250/600/1200\,\mu$m per head/layer/seed/patient. A coordinate-selected remote landmark feature was zeroed to test functional dependency, not biological correlation. The degree/length control accepted no swaps in 64/66 recovered-cohort patient–seed graphs, with at most one swap in the remaining graphs; those cases supply little independent comparison.

AI-assisted computational tools. Truong-Son Hy established the scientific ideas, objectives, research direction and experimental instructions and provided intellectual oversight. Under the author's direction, OpenAI Codex assisted with coding, data processing, model training/execution and computational tests; ChatGPT assisted with drafting, editing and La-

TeX preparation. The author is responsible for evaluating the analyses, their interpretation and the conclusions. AI tools are not authors.

5 Ethics

This work reanalyzed publicly released, anonymized datasets without new recruitment, tissue collection or experimental intervention. The original studies describe their ethics and consent procedures[2–4].

6 Data availability

Development data were drawn from HEST[5]. HER2ST data are available at https://zenodo.org/records/4751624, Wu data at https://zenodo.org/records/4739739, and Bassiouni data under GEO accession GSE210616, https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE210616. The selected Bassiouni inputs have been recovered and verified against original checksums. Source tables for all manuscript figures and tables accompany the journal package. The complete additional per-gene exports (approximately 523 MB) and newly trained model-state bundle (approximately 102 MB) are currently author-local, with file/checksum manifests in the repository; no public deposit or DOI is claimed. Raw tissue images and complete prediction caches are not redistributed with the manuscript.

7 Code availability

Code, protocols and reproduction instructions are available at https://github.com/HySonLab/GraphPathology. Publication figures and documents can be regenerated from the recorded exports. Full replay of the original image predictions remains limited by three missing historical primary image-head binaries. The additional 30 and 48 fitted states are distinct from those missing binaries; their release-ready manifests specify author-local access and required deposition actions.

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10 Author contributions

Truong-Son Hy was responsible for study conception and design, methodology, intellectual oversight, evaluation of analyses and validation, interpretation, visualization, and manuscript writing and revision. Computational and editorial tool assistance is disclosed in Methods.

11 Competing interests

Truong-Son Hy is an Editorial Board Member of *Scientific Reports*. The author declares no other competing interests.

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